



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

ARYAN PATIL 25070122051 CSE-A2

Subject: Programming with Java

ASSIGNMENT No :3

TITLE :Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Problem Statement

Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Learning Objectives

To implement method overloading and use static variables and methods in Java programs.

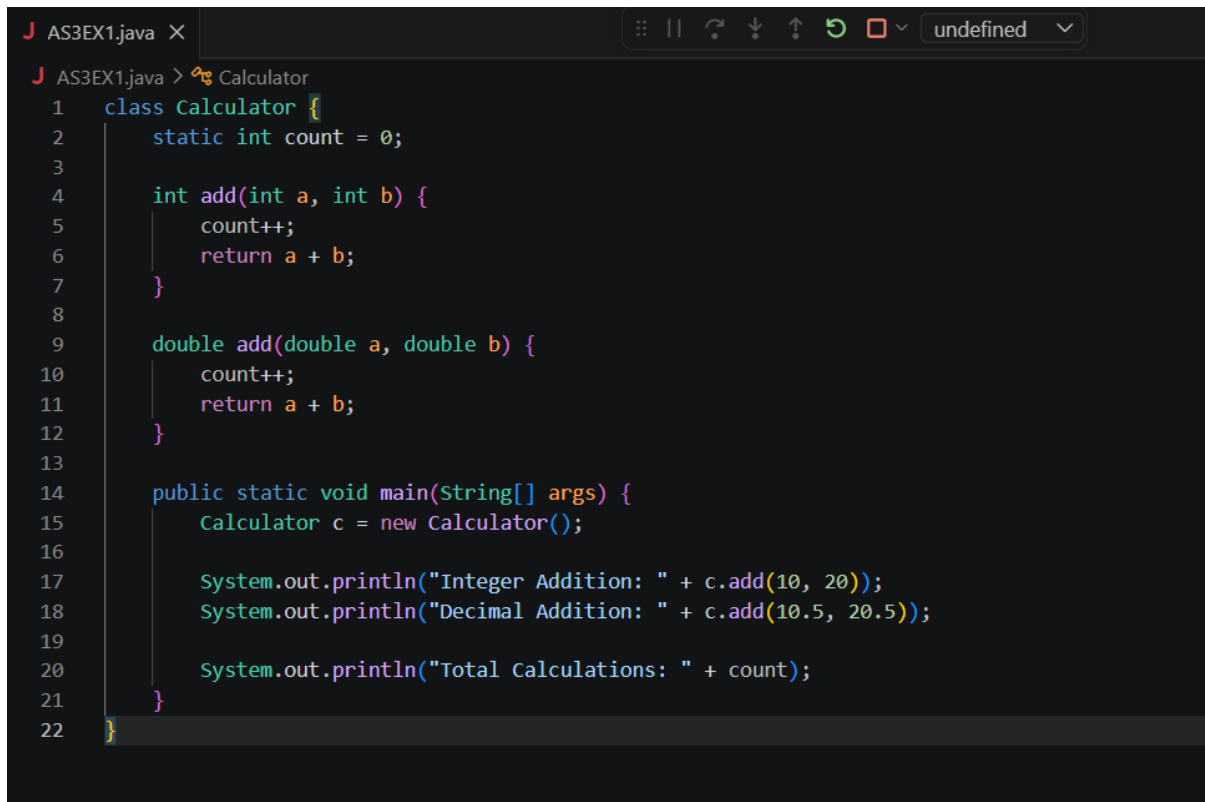
Theory

Method overloading allows multiple methods to have the same name with different parameter lists, enabling compile-time polymorphism. The static keyword is used for variables and methods that belong to the class instead of individual objects. Static members share common data among all objects and reduce memory usage. These concepts improve code flexibility, readability, and efficient resource management.

CONCLUSION- This assignment demonstrated the implementation of method overloading and the use of static variables and methods in Java. Method overloading provided flexibility by allowing multiple methods with the same name to perform similar operations using different parameter lists, while static members enabled data and methods to be shared across all objects of a class. The Calculator application showed how overloaded methods can handle different data types and how a static variable can track the number of calculations. Similarly, the Restaurant Billing Application demonstrated the use of overloaded methods for different order types and static variables to maintain the total number of orders. Overall, this assignment strengthened the understanding of compile-time polymorphism and efficient class design in Java.

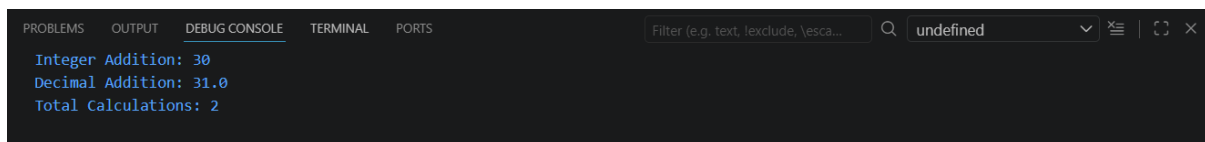
EXCERCISES-

1. Develop a **Calculator program** using overloaded methods for addition of integers and decimals. Use a static variable to count calculations.



```
AS3EX1.java X
AS3EX1.java > Calculator
1 class Calculator {
2     static int count = 0;
3
4     int add(int a, int b) {
5         count++;
6         return a + b;
7     }
8
9     double add(double a, double b) {
10        count++;
11        return a + b;
12    }
13
14    public static void main(String[] args) {
15        Calculator c = new Calculator();
16
17        System.out.println("Integer Addition: " + c.add(10, 20));
18        System.out.println("Decimal Addition: " + c.add(10.5, 20.5));
19
20        System.out.println("Total Calculations: " + count);
21    }
22 }
```

OUTCOME-



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Integer Addition: 30
Decimal Addition: 31.0
Total Calculations: 2
```

2. Develop a **Restaurant Billing Application** where overloaded methods calculate bills for dine-in, takeaway, and delivery orders, while static variables track total orders.

```
J AS3EX1.java J AS3EX2.java X
J AS3EX2.java > Restaurant
1 class Restaurant {
2     static int totalOrders = 0;
3
4     // Dine-in bill
5     void bill(double amount) {
6         totalOrders++;
7         System.out.println("Dine-In Bill: Rs." + amount);
8     }
9
10    // Takeaway bill
11    void bill(double amount, double packingCharge) {
12        totalOrders++;
13        System.out.println("Takeaway Bill: Rs." + (amount + packingCharge));
14    }
15
16    // Delivery bill
17    void bill(double amount, double packingCharge, double deliveryCharge) {
18        totalOrders++;
19        System.out.println("Delivery Bill: Rs." +
20            (amount + packingCharge + deliveryCharge));
21    }
22
23    Run main | Debug main
24    public static void main(String[] args) {
25        Restaurant r = new Restaurant();
26
27        r.bill(500);           // Dine-in
28        r.bill(500, 20);      // Takeaway
29        r.bill(500, 20, 50);  // Delivery
30
31        System.out.println("Total Orders: " + totalOrders);
32    }
}
```

OUTCOME-

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Dine-In Bill: Rs.500.0
Takeaway Bill: Rs.520.0
Delivery Bill: Rs.570.0
Total Orders: 3
```